

# A COMPACT BUCK CONVERTER

## Suitable For Wearables And IoT

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A buck converter is a DC-DC power device that efficiently steps down a voltage to a lower level. It uses an inductor, resistors, and a capacitor to regulate the output, making it particularly suitable for wearable electronics and IoT devices.

Low-dropout regulators (LDOs) offer a more straightforward approach to voltage regulation, but they often dissipate significant energy as heat, thereby undermining efficiency in compact, power-sensitive systems. Buck converters provide a more efficient solution, although the inclusion of an inductor increases the design footprint.

This compact converter design is based on a buck converter IC. The complete power system, including the converter, fits within a 9mm footprint—ideal for space-constrained applications while leaving room for other essential components.

A key feature is its ability to deliver variable output voltages ranging from 1.2V to 3.3V DC, achieved by replacing a fixed resistor with a potentiometer. This enables fine-tuned voltage adjustment to match the specific needs of each device.

The converter may be used as a standalone variable-voltage module or integrated directly into wearable electronics.

The converter is embedded within the device's PCB, providing the desired output voltage range with over 93% efficiency and no thermal issues. The design incorporates the PAM2305D SMD IC, housed in a

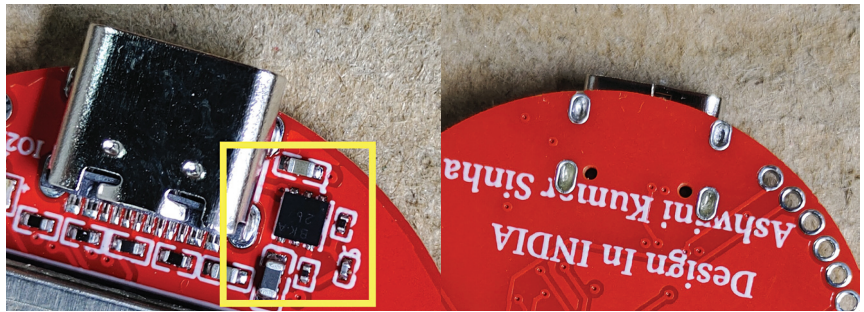


Fig. 1: Author's DC-DC buck converter integrated into the custom-designed PCB

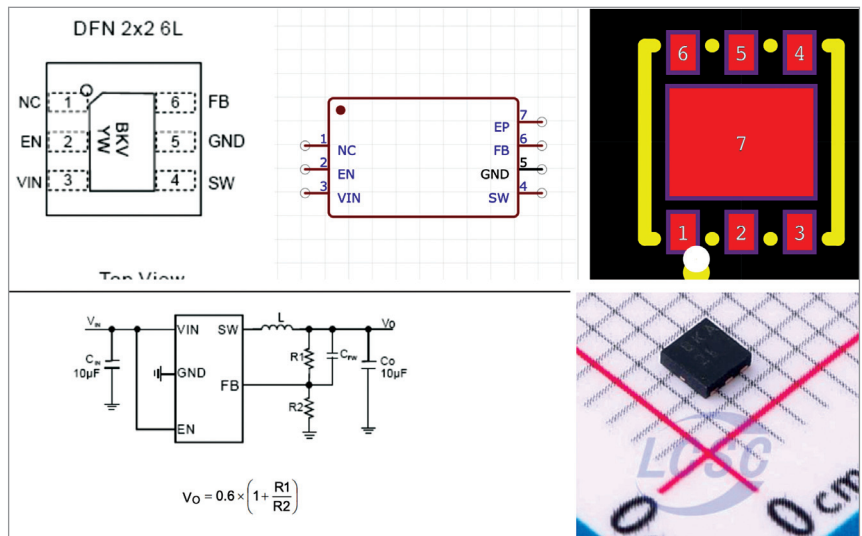


Fig. 2: IC datasheet pinout and application circuit

TABLE 1: BILL OF MATERIALS

| Name                 | Designator | Footprint                           | Quantity | Manufacturer Part  | Manufacturer | Supplier | Supplier Part |
|----------------------|------------|-------------------------------------|----------|--------------------|--------------|----------|---------------|
| 10µF Capacitor       | C1, C2     | C0402                               | 2        | CL05A106MP5NUNC    | SAMSUNG      | LCSC     | C315248       |
| 100pF Capacitor      | C3         | C0201                               | 1        | GRM0335C1E101JA01D | muRata       | LCSC     | C76917        |
| 4.7µH Inductor       | L1         | L0603                               | 1        | CMH160808B4R7MT    | FH           | LCSC     | C394952       |
| 500kΩ Resistor       | R1         | RES-ADJ-SMD_3P-L3.8-W3.7_VGF39NCHXT | 1        | VGF39NCHXT-B504    | HDK          | LCSC     | C22461638     |
| 120kΩ Resistor       | R2         | R0201                               | 1        | 0201WMF1203TEE     | UNI-ROYAL    | LCSC     | C270362       |
| USB 2.0 Solder Pad   | U1         | USB SOLDER PADS                     | 1        | USB-TYPE-C-018     | DEALON       | LCSC     | C2927038      |
| PAM2305DGFADJ        | U8         | DFN-6_L2.0-W2.0-P0.65-BL-EP         | 1        | PAM2305DGFADJ      | DIODES       | LCSC     | C780869       |
| Potmeter SMD (Fig 5) | R1 (Fig 5) | RES-ADJ-SMD_3P-L3.8-W3.7_VGF39NCHXT | 1        | VGF39NCHXT-B504    | HDK          | LCSC     | C22461638     |